Homework 8

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MATH 421 — The Theory of Single Variable Calculus
Section 003
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Collaboration Statement

Problem 1 *(Cauchy Sequence: $s_n = \frac{1}{n}$)*

Prove, directly from the definitions, that (s_n) where $s_n = \frac{1}{n}$ is a Cauchy sequence.

Discussion.

Proof.

$\square$

Problem 2 *(Cauchy Sequence: $x_n = \sqrt{n}$)*

Let $x_n = \sqrt{n}$. Show that while the difference between consecutve terms goes to 0, i.e., show $\lim |x_{n+1} - x_n| = 0$, the sequence (x_n) is not a Cauchy sequence.

Discussion.

Proof.

$\square$

Problem 3 *(Ross Chapter 2 10.6 (a))*

Let (s_n) be a sequence such that $|s_{n+1} - s_n| < 2^{-n}$ for all $n \in \mathbb{N}$. Prove that (s_n) is a Cauchy sequence and hence a cin

Discussion.

Proof.

□

Problem 4 *(Ross Chapter 2 14.6)*

Discussion.

Proof.

□

Problem 5 *(Ross Chapter 2 14.7)*

Discussion.

Proof.

□

Problem 6 *($\varepsilon - \delta$ Convgernce Proof)*

Prove using the $\varepsilon - \delta$ definition of the limit of a function that if $\lim_{x\to a} f(x) = L$, then $\lim_{x\to a} 4f(x) = 4L$.

Discussion.

Proof.

□